IMEN574 / IDSC574 · PROGRAMMING FOR AX

8

Regression

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Spring 2027

Where We Are

01 Programming with AI

Lectures 2–4

02 **Python and Classical Machine Learning**

Lectures 5–13

03 Deep Learning and Foundation Models

Lectures 14–22

Outline: Regression

01 **Fit a Numerical Target**

Connect loss to learned parameters

02 **Control the Function Family**

Choose a useful inductive bias

03 **Test Nonlinear Alternatives**

Measure behavior under one protocol

01 · Fit a Numerical Target

01 **Fit a Numerical Target**

Connect loss to learned parameters

02 **Control the Function Family**

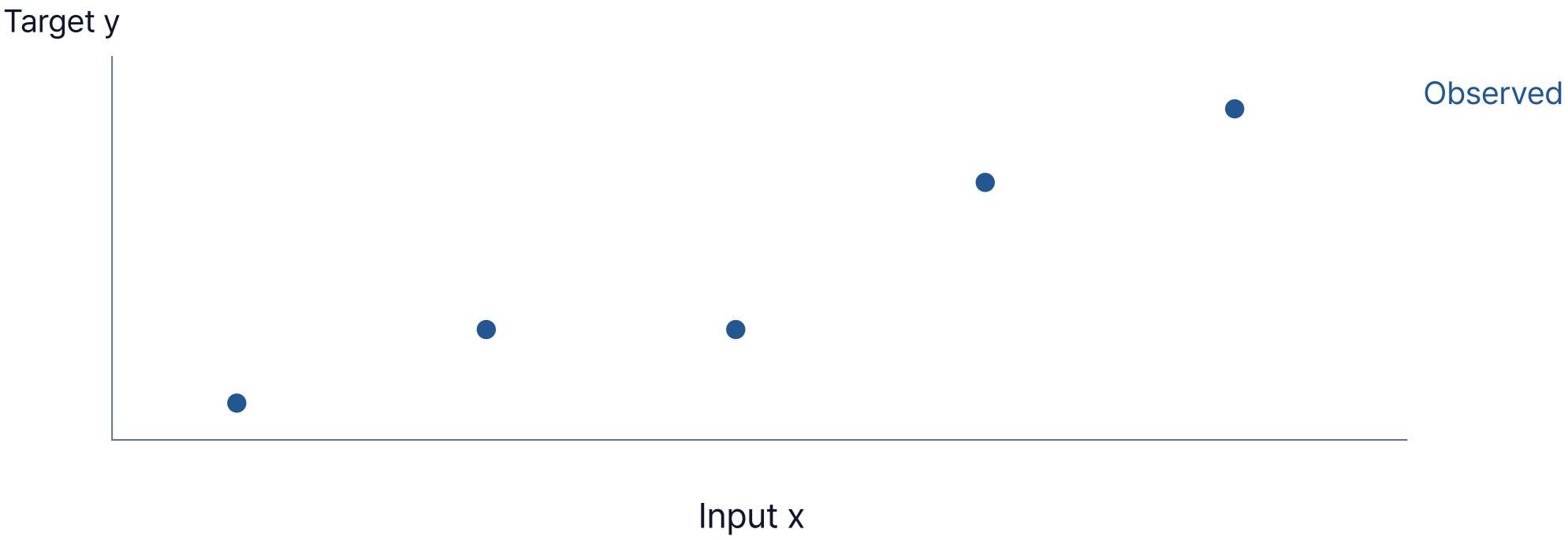
Choose a useful inductive bias

03 **Test Nonlinear Alternatives**

Measure behavior under one protocol

Why a Regression Model

A numerical target needs a rule for unseen inputs



Five observations do not determine one unique function

The Regression Task

Predict a quantity from an input vector



Available measurements

Fitted from data

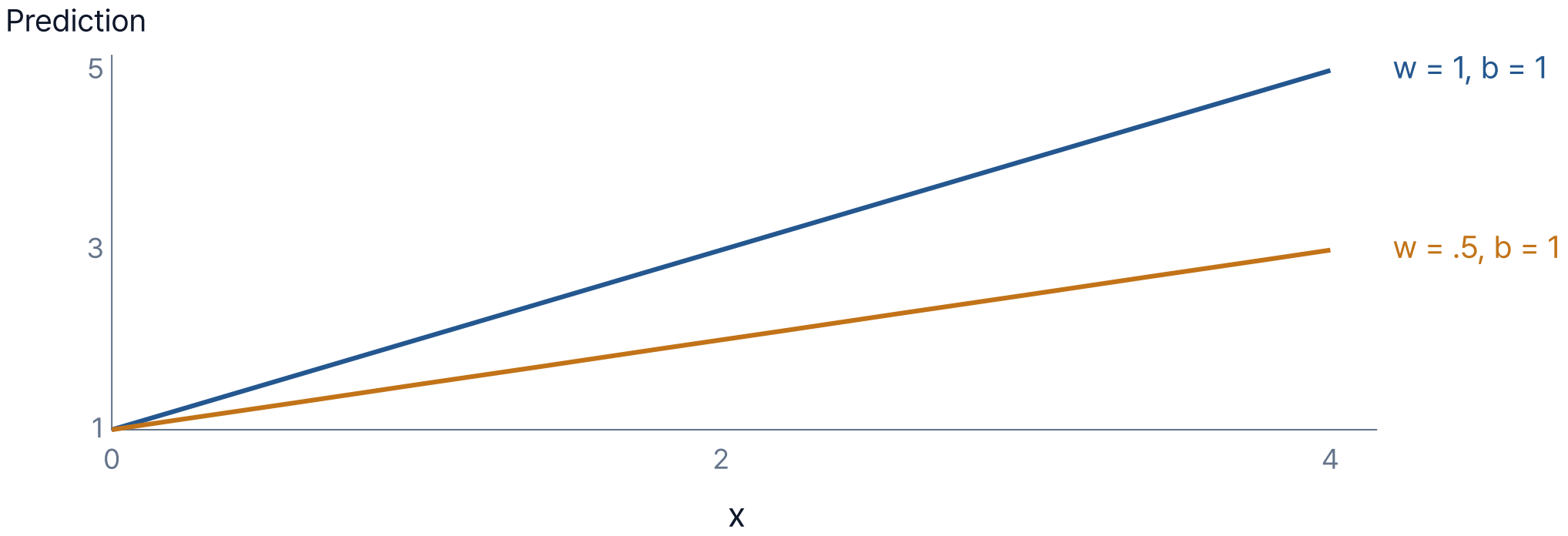
One target value

The task is defined by the target meaning

Function and Parameters

A model family limits the functions we can fit

$$f(x; w, b) = wx + b$$



Two parameters select one line from the family

Residuals

Compare each prediction with its observed target

x	Target	Prediction	Residual
0	1	1	0
1	2	2	0
2	2	3	-1
3	4	4	0
4	5	5	0

The default line overpredicts one observed target

Squared Error

Penalize errors without cancelling their signs

$$\text{MSE} = \text{mean}((y - f(x))^2)$$



The fitted intercept lowers this toy squared error

A Mean Baseline

Compare the line with a constant prediction

```
import numpy as np
y = np.array([1.,2.,2.,4.,5.])
pred = np.repeat(y.mean(), len(y))
print(y.mean())
print(np.mean((y-pred)**2))
```

```
2.8
2.16
```

The baseline ignores x but still minimizes constant-model MSE

Least-Squares Solution

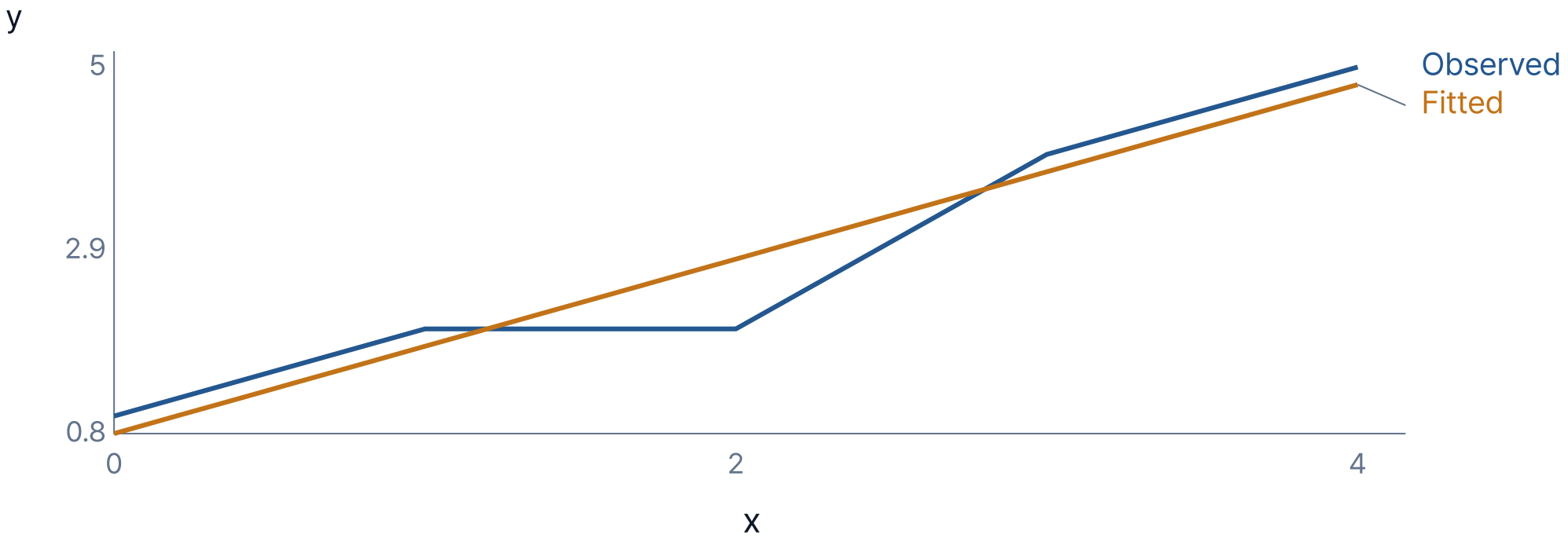
Use a stable solver instead of forming an inverse

```
X = np.column_stack([x, np.ones(len(x))])
coef, *_ = np.linalg.lstsq(X, y, rcond=None)
w, b = coef
pred = X @ coef
```

A solver computes the fitted line without an explicit inverse

The Fitted Line

The intercept is .8 for these five observations

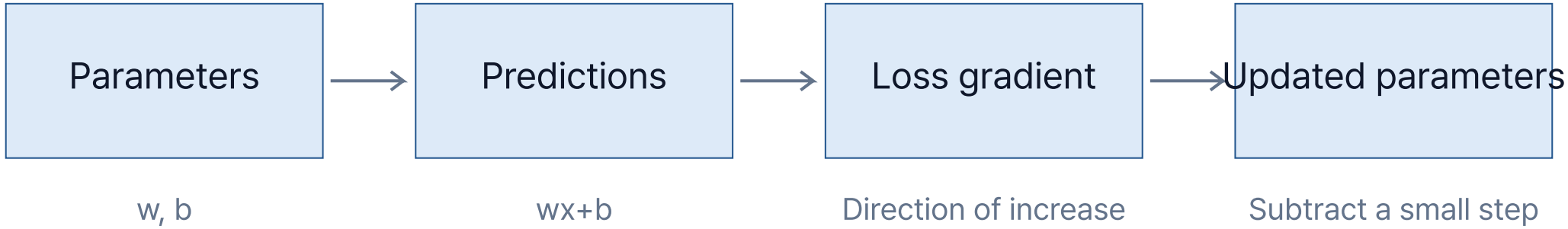


$$f(x) = 1.0x + 0.8$$

The fit distributes residuals across the observations

Gradient Descent

Move parameters against the loss gradient



The update follows a computed direction

Gradient Update Rule

The current residuals determine the next update

$$dw = 2\ \mathrm{mean}(x(wx+b-y))$$
$$db = 2\ \mathrm{mean}(wx+b-y)$$

One Gradient Step

The current residuals determine the next update

	w	b
Before	1.00	1.00
Gradient	0.80	0.40
After η=.1	0.92	0.96

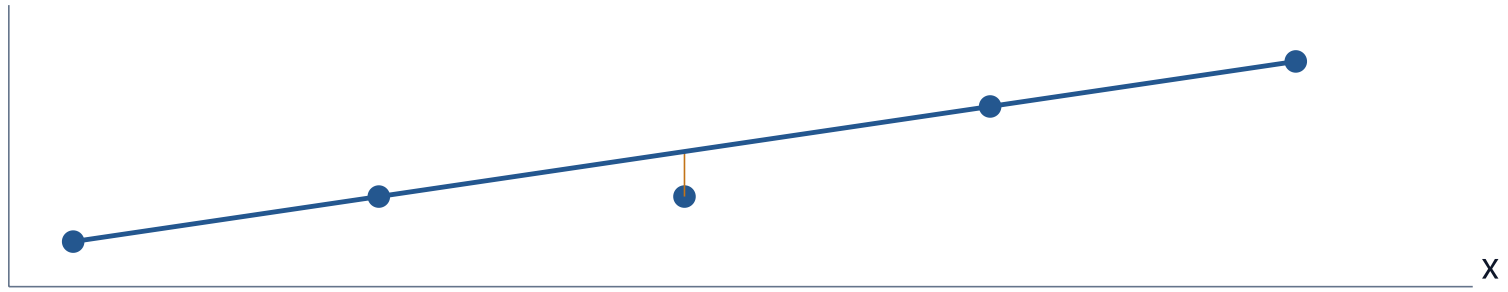
A gradient step updates both parameters

Demo: Fitting a Line

Change slope and intercept and inspect the residuals

Adjust slope and intercept; compare the squared errors

Slope 1 Intercept 1



ŷ = 1.00x + 1.00 | MSE 0.200
Least squares: slope 1.0, intercept 0.8, MSE 0.16

Answer: Least-Squares Fit

The optimal intercept spreads the residuals

$$w = 1,\ b = .8$$

$$\text{residuals} = [.2,\ .2,\ -.8,\ .2,\ .2]$$

$$\text{MSE} = .16$$

The fitted residuals satisfy the line-model normal conditions

02 · Control the Function Family

01 **Fit a Numerical Target**

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Why Regularization

A flexible fit can follow noise rather than signal

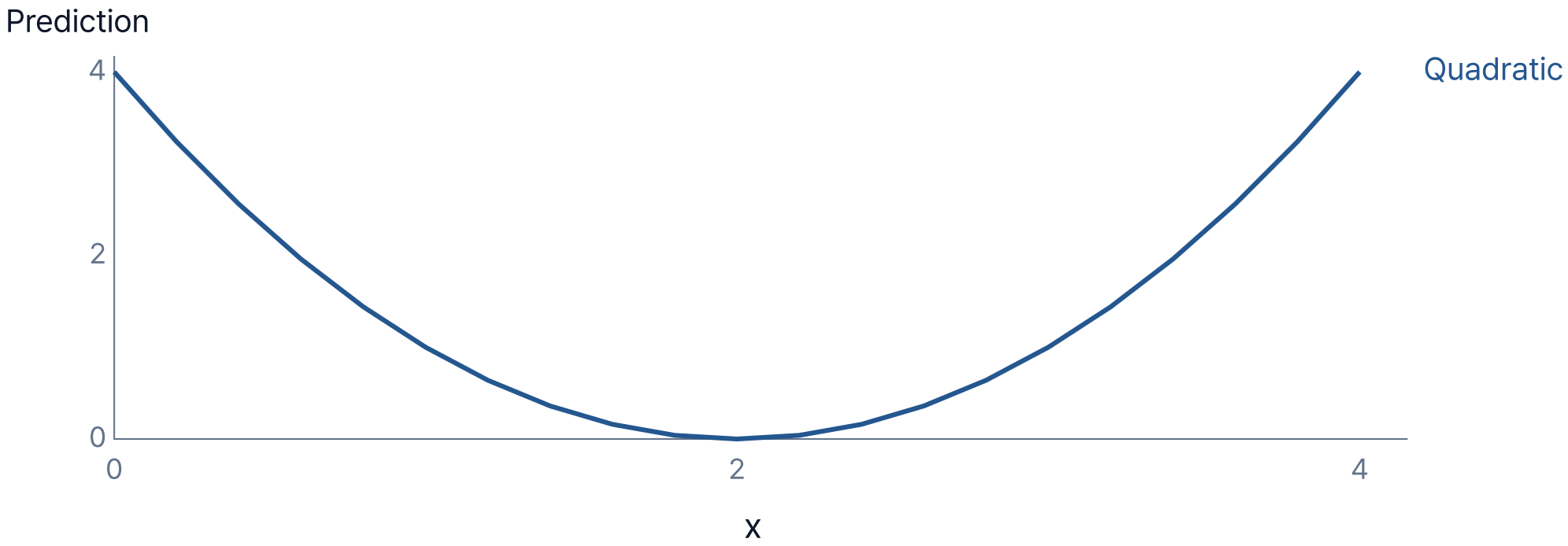
SMALL FAMILY	LARGE FAMILY
Few parameters	Many parameters
Limited training fit	Can fit training details
Stable across small data changes	May change strongly with the sample

Model flexibility changes what can be fitted

Polynomial Features

Linearity in parameters can describe a curved function

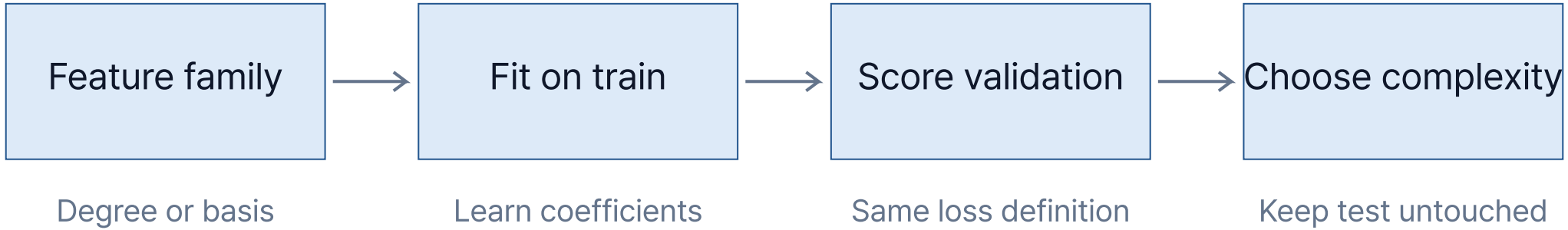
$$f(x) = b + w_1 x + w_2 x^2$$



A linear parameter model can use nonlinear features

Training and Validation Error

Choose complexity using held-out behavior

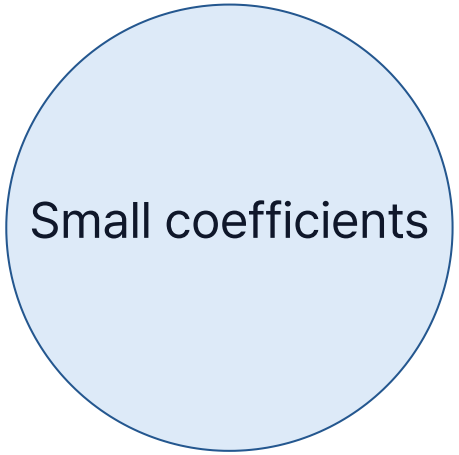


The validation set judges the chosen flexibility

Ridge Regression

Penalize coefficient size along with prediction error

$$\min \text{mean}((y - Xw - b)^2) + \lambda ||w||^2$$



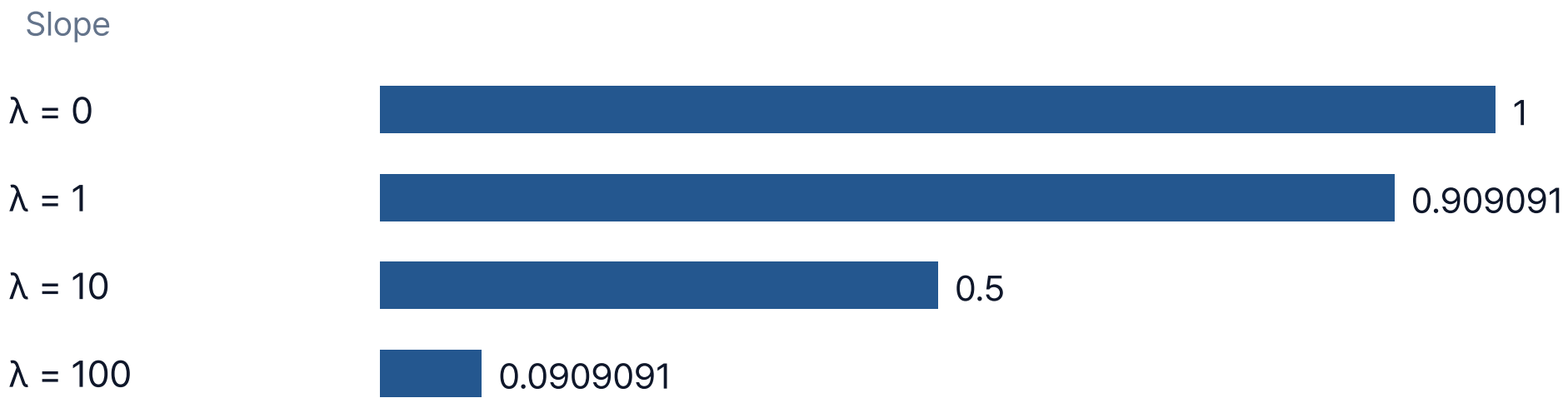
The intercept is not penalized here

The penalty discourages unnecessarily large coefficients

https://scikit-learn.org/stable/modules/linear_model.html

Shrinkage on a Toy Input

For the sum-loss convention, w = 10 / (10 + λ)



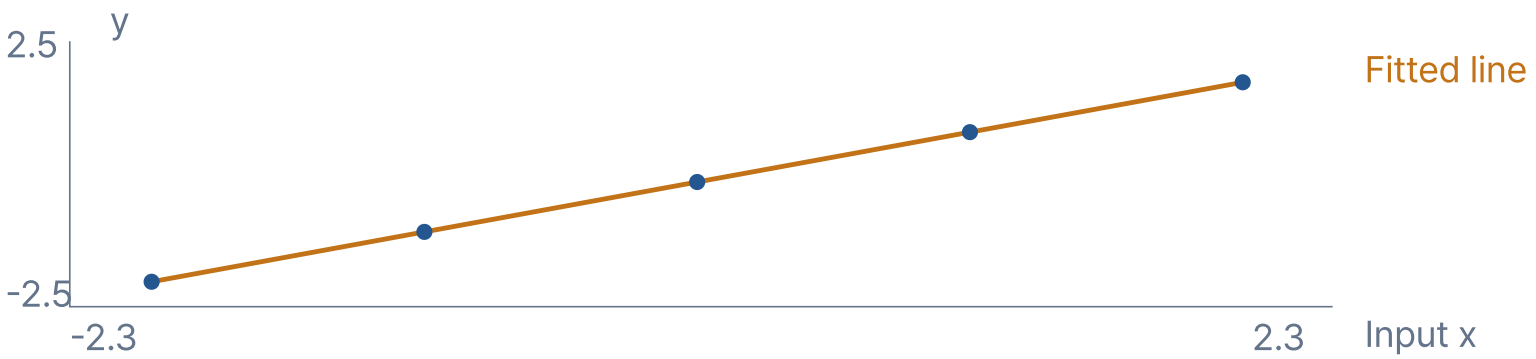
The slope shrinks as the stated sum-loss penalty grows

Demo: Ridge Shrinkage

A stronger coefficient penalty changes fit and residuals

Change lambda from 0 to 10 Predict the slope and training squared error

Lambda (sum loss) 0



w=10/(10+lambda)=1; SSE=0

Penalty=0; sum-loss objective=0

Answer: Ridge Shrinkage

The same input produces these calculated values

Input / quantity	Result
sum x^2=10; y=x; lambda0	slope1; SSE0
lambda10	slope0.5; SSE2.5
lambda100	slope0.090909; SSE8.264463

Caveat: Ridge Shrinkage

Penalty scale depends on the loss convention

Condition	Consequence
sum squared loss + lambda*w^2	w=10/(10+lambda)
mean squared loss + lambda*w^2	w=2/(2+lambda)

Caveat: Feature Scale

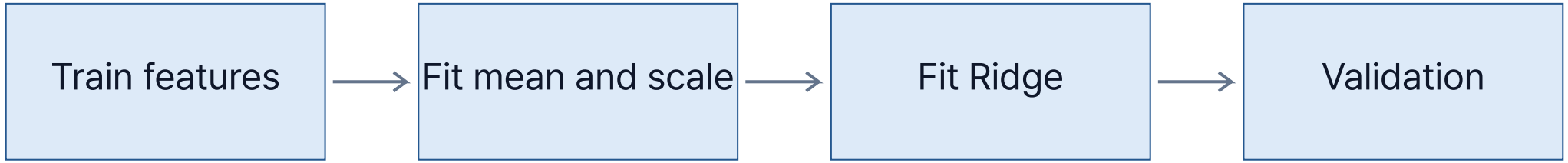
The same penalty acts differently across units

Representation	Feature Scale	Equivalent Coefficient
Length in m	1	w
Length in mm	1000	w / 1000

Units change coefficient sizes without changing predictions

Scaling Inside the Pipeline

Learn the scale from training data only



Declared units

Stored statistics

Choose λ on validation

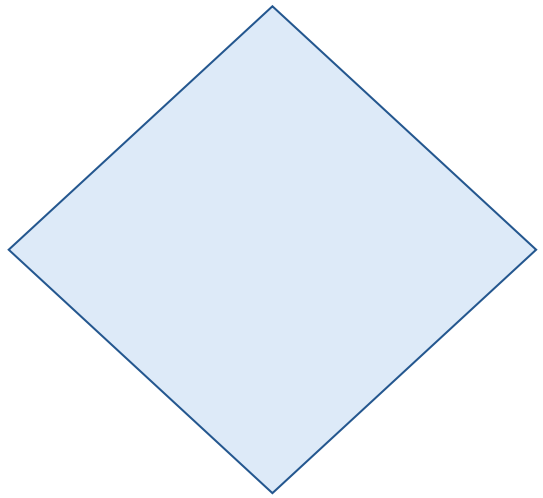
Reuse the statistics

The penalty comparison needs a consistent feature scale

Lasso Regression

An L1 penalty can set some coefficients to zero

$$\min \text{loss} + \lambda \sum |w_j|$$



L1 constraint geometry

Corners on coordinate axes

The L1 constraint has corners on coordinate axes

https://scikit-learn.org/stable/modules/linear_model.html

Ridge and Lasso Choices

The penalty expresses a preference for the solution

Question	Ridge	Lasso
Coefficient behavior	Shrink together	May become zero
Correlated features	Shares weight	May select one
Selection goal	Stability	Sparse representation

The choice depends on the desired solution and validation

Coefficient Interpretation

Condition on the other included features

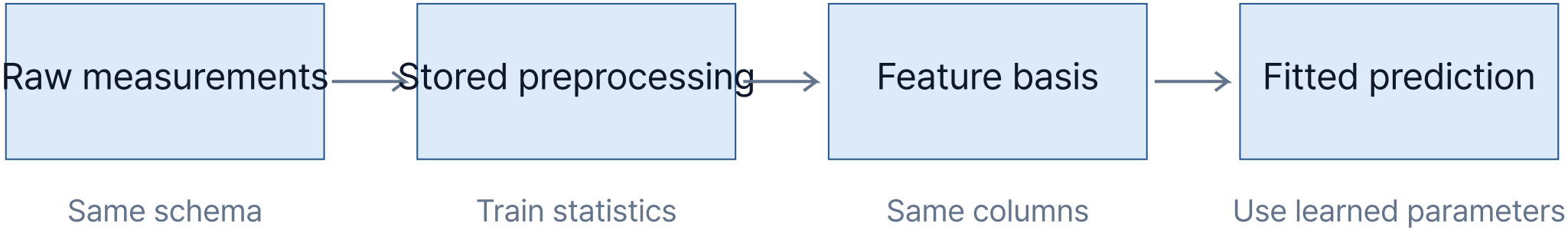
```
LINEAR INTERPRETATION

A one-unit increase in x changes prediction by w
Other included features are held fixed
The unit and preprocessing must be stated
Association does not establish a causal intervention
```

A coefficient describes the fitted model under its feature definition

A Learned Pipeline

Save transformations with the fitted coefficients



The prediction depends on more than the final coefficient vector

03 · Test Nonlinear Alternatives

01 **Fit a Numerical Target**

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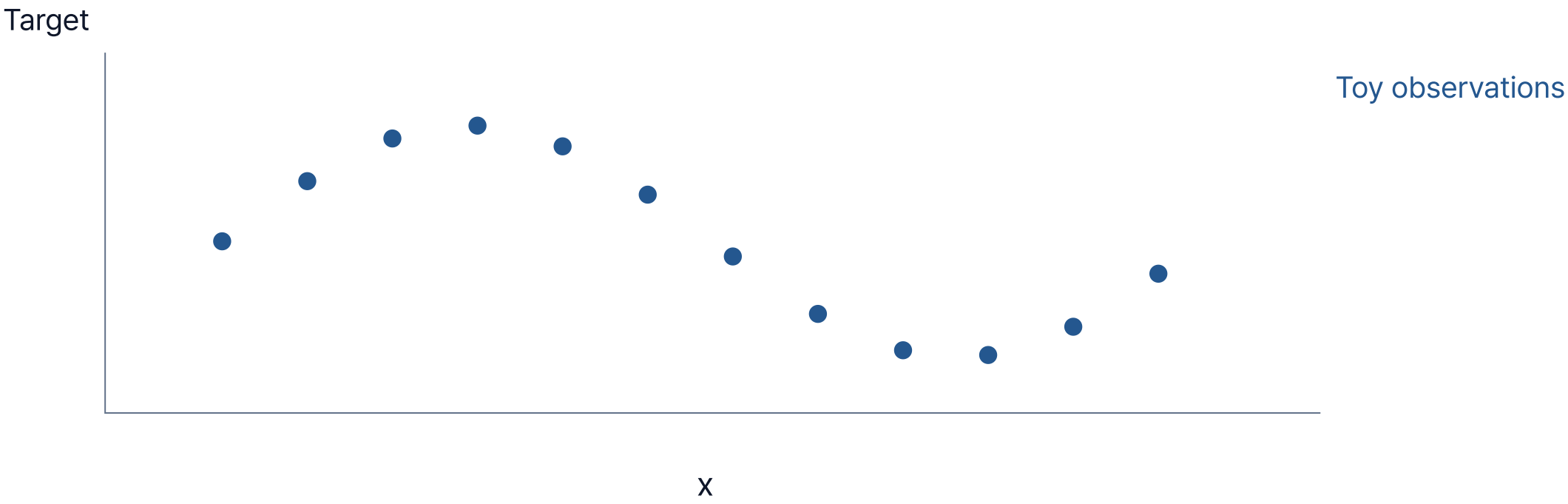
Choose a useful inductive bias

03 **Test Nonlinear Alternatives**

Measure behavior under one protocol

Why Nonlinear Similarity

Distant inputs need not share one linear trend

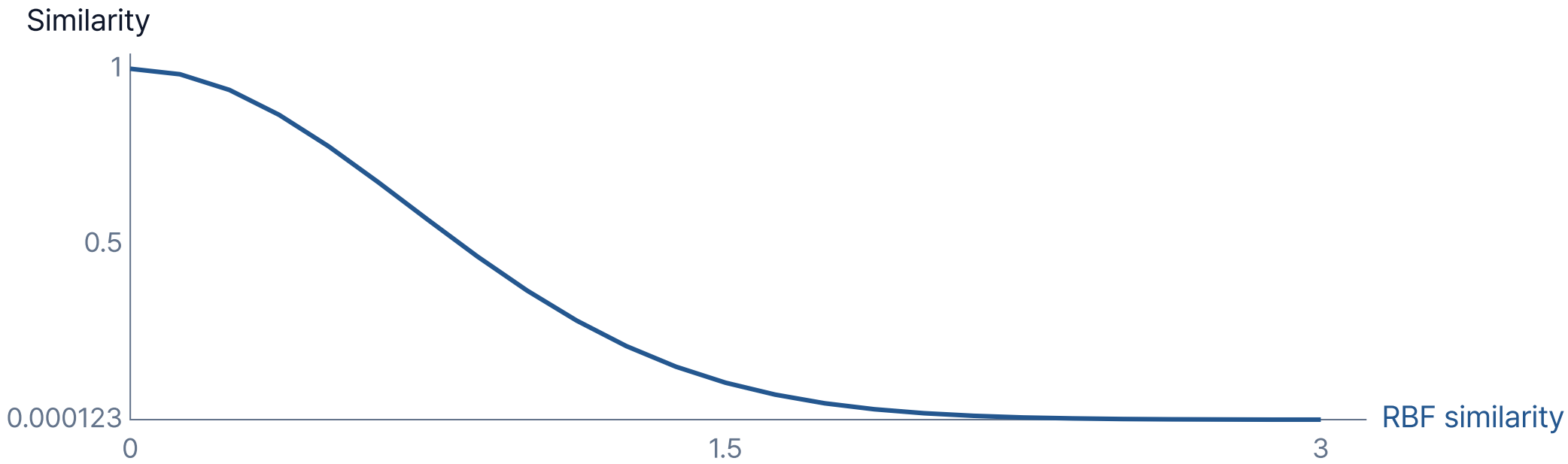


The toy relationship bends beyond a single straight line

Kernel Similarity

A kernel compares inputs through a feature representation

$$K(x, x') = \exp(-\gamma \, ||x - x'||^2)$$



This RBF kernel gives nearby points greater similarity

The Kernel Matrix

Compare every training input with every other input

	x=0	x=1	x=2
x=0	1.00	.37	.02
x=1	.37	1.00	.37
x=2	.02	.37	1.00

Pairwise distances determine this symmetric similarity matrix

Kernel Ridge Solution

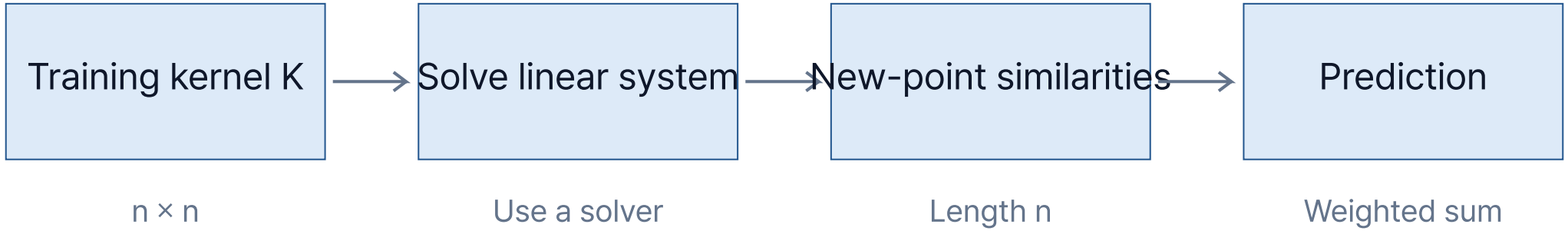
Solve for weights on training-input similarities

$$\alpha = (K + \lambda I)^{-1} y$$

$$f(x) = \sum_i \alpha_i K(x, x_i)$$

Kernel Ridge Regression

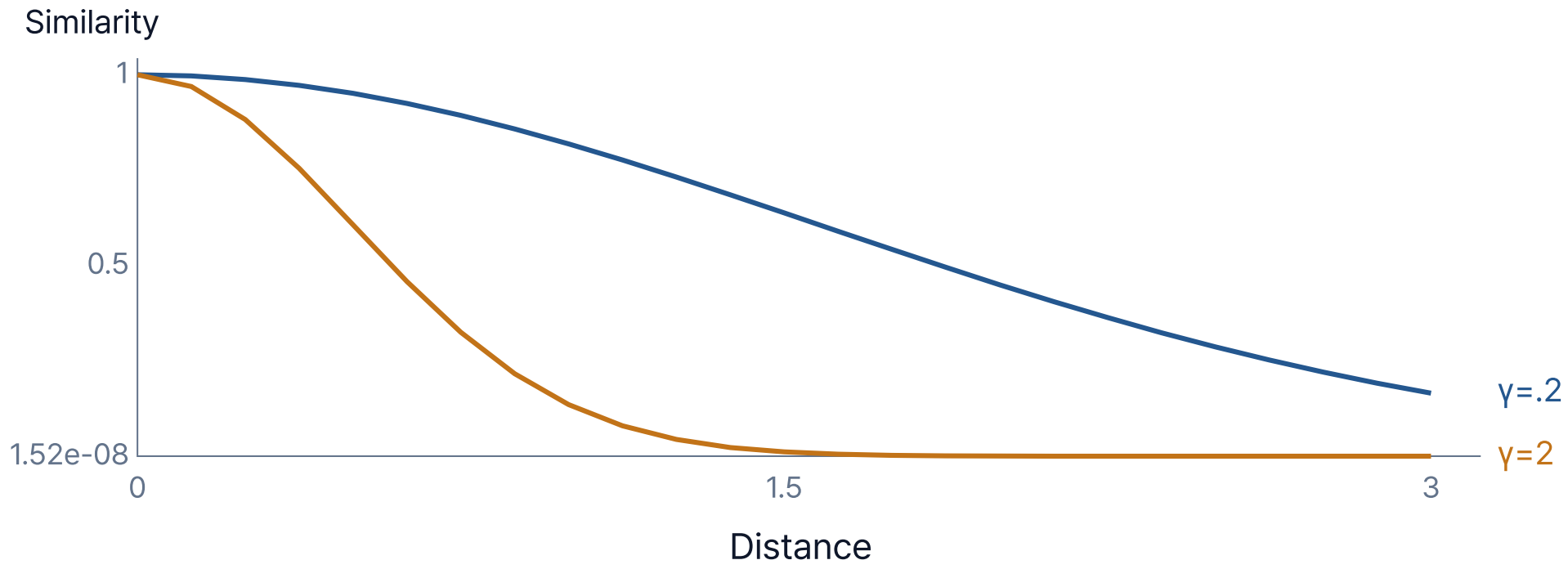
Solve for weights on training-input similarities



The learned weights combine training-point similarities

Kernel Width

Change the distance scale that counts as nearby



A larger gamma narrows the region of similarity

A Fixed Validation Split

Compare every model on the same four held-out inputs

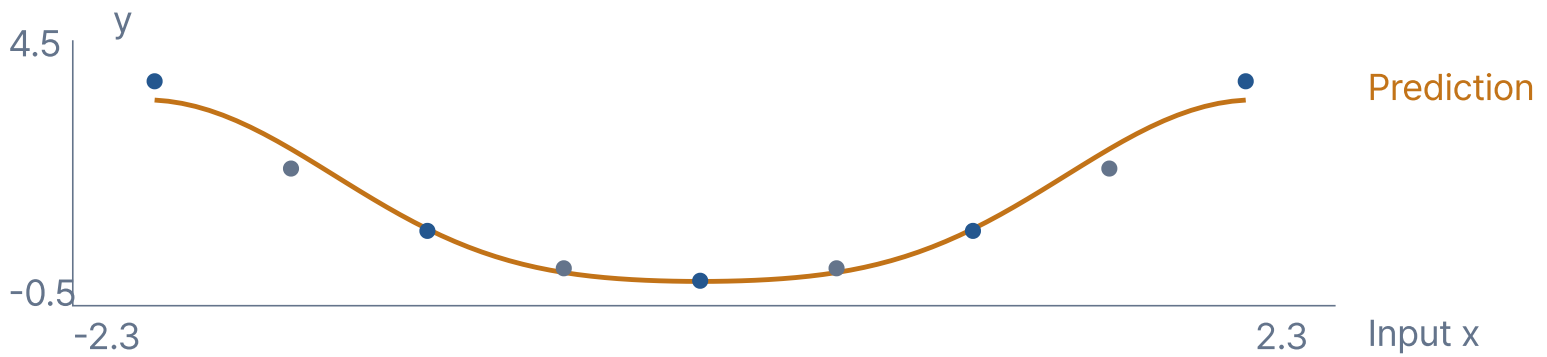
Partition	x	Target $y=x^2$
Train	−2, −1, 0, 1, 2	4, 1, 0, 1, 4
Validation	−1.5, −0.5, 0.5, 1.5	2.25, 0.25, 0.25, 2.25
Linear ridge	Centered slope = 0	Intercept = train mean = 2

Demo: Kernel Validation

Compare training fit with untouched validation data

Compare linear ridge and RBF kernel ridge Change gamma and lambda; compare train and validation MSE

Model rbf Lambda 0.1 RBF gamma 1



Train MSE 0.057317; validation MSE 0.080523
Validation predictions [2.6422, 0.1651, 0.1651, 2.6422]

Answer: Kernel Validation

The same input produces these calculated values

Model / settings	Train MSE	Validation MSE
RBF gamma1 lambda0.1	0.057317	0.080523
RBF gamma10 lambda0.1	0.056196	1.776762
Linear ridge lambda0.1	2.8	1.5625

Caveat: Kernel Validation

Kernel width controls interpolation and extrapolation

Condition	Consequence
Very narrow kernels	Small train loss can coexist with poor validation loss
Input far from every training point	This zero-mean RBF model tends toward zero

Caveat: Extrapolation

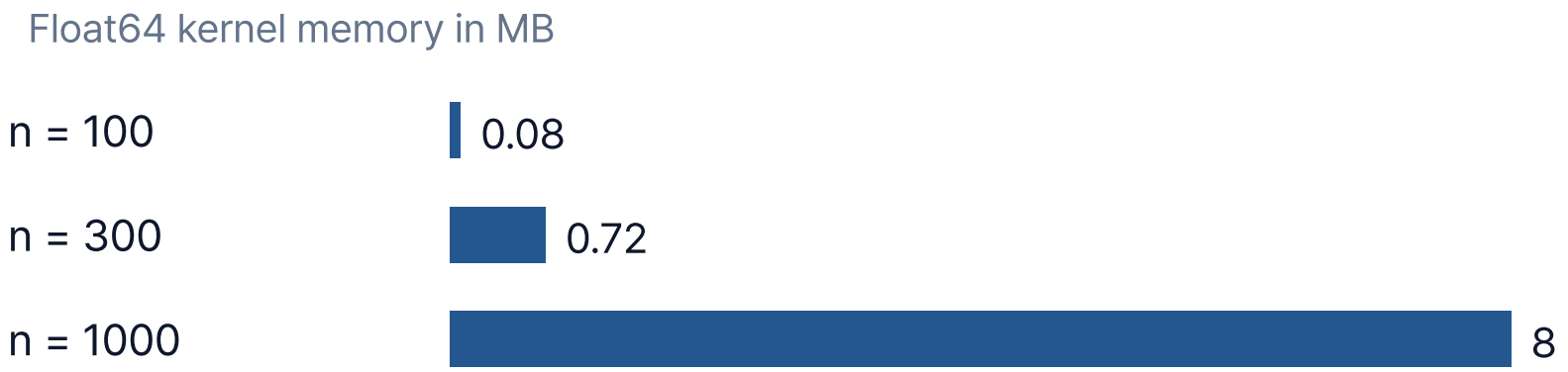
A good fit inside the data does not verify outside behavior

OBSERVED DOMAIN	UNSEEN DOMAIN
Training x spans 0 to 4	Deployment asks for x = 10
Validation uses the same range	No nearby observations
Interpolation is tested	Extrapolation remains untested

The validation range limits the claim

Kernel Computation Cost

A dense training kernel grows quadratically in storage



Ten times as many rows require one hundred times as many kernel entries

A Holdout Comparison

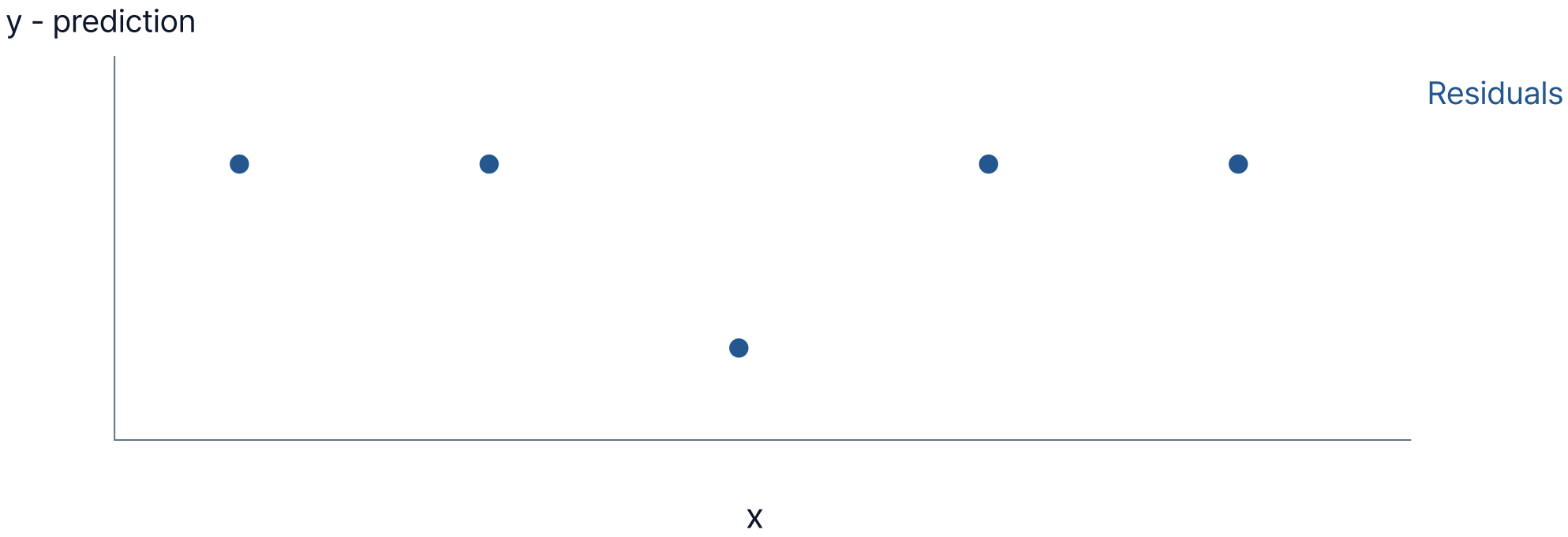
Compare all candidates under the same split and loss

Candidate	Holdout	Metrics
Mean baseline	Same split	MAE, RMSE
Linear / Ridge	Same split	MAE, RMSE
Kernel Ridge	Same split	MAE, RMSE

The comparison fixes the evaluation conditions

Residual Diagnostics

Inspect where the errors remain



The largest residual occurs at the middle observation

A Model Choice Record

State the selected family and the evidence

```
MODEL RECORD

Target, units and feature availability
Fixed split and preprocessing
Validation loss for each candidate
Selected settings and final test score
Residual patterns and untested ranges
```

The chosen model is a conditional decision

Takeaway: Regression

01 A fitted model minimizes a chosen loss within a family

02 Regularization and scale jointly shape the solution

03 Compare candidates on the same unseen data